Pinto bean consumption reduces biomarkers for heart disease risk.

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OBJECTIVE: To determine effects of daily intake of 1/2 cup pinto beans, black-eyed peas or carrots (placebo) on risk factors for coronary heart disease (CHD) and diabetes mellitus (DM) in free-living, mildly insulin resistant adults over an 8 week period. METHODS: Randomized, crossover 3x3 block design. Sixteen participants (7 men, 9 women) received each treatment for eight-weeks with two-week washouts. Fasting blood samples collected at beginning and end of periods were analyzed for total cholesterol (TC), low density lipoprotein cholesterol (LDL-C), high density lipoprotein cholesterol, triacylglycerols, high-sensitivity C-reactive protein, insulin, glucose, and hemoglobin A1c. RESULTS: A significant treatment-by-time effect impacted serum TC ($p = 0.026$) and LDL ($p = 0.033$) after eight weeks. Paired t-tests indicated that pinto beans were responsible for this effect ($p = 0.003$; $p = 0.008$). Mean change of serum TC for pinto bean, black-eyed pea and placebo were -19 +/- 5, 2.5 +/- 6, and 1 +/- 5 mg/dL, respectively ($p = 0.011$). Mean change of serum LDL-C for pinto bean, black-eyed pea and placebo were -14 +/- 4, 4 +/- 5, and 1 +/- 4 mg/dL, in that order ($p = 0.013$). Pinto beans differed significantly from placebo ($p = 0.021$). No significant differences were seen with other blood concentrations across the 3 treatment periods. CONCLUSIONS: Pinto bean intake should be encouraged to lower serum TC and LDL-C, thereby reducing risk for CHD.